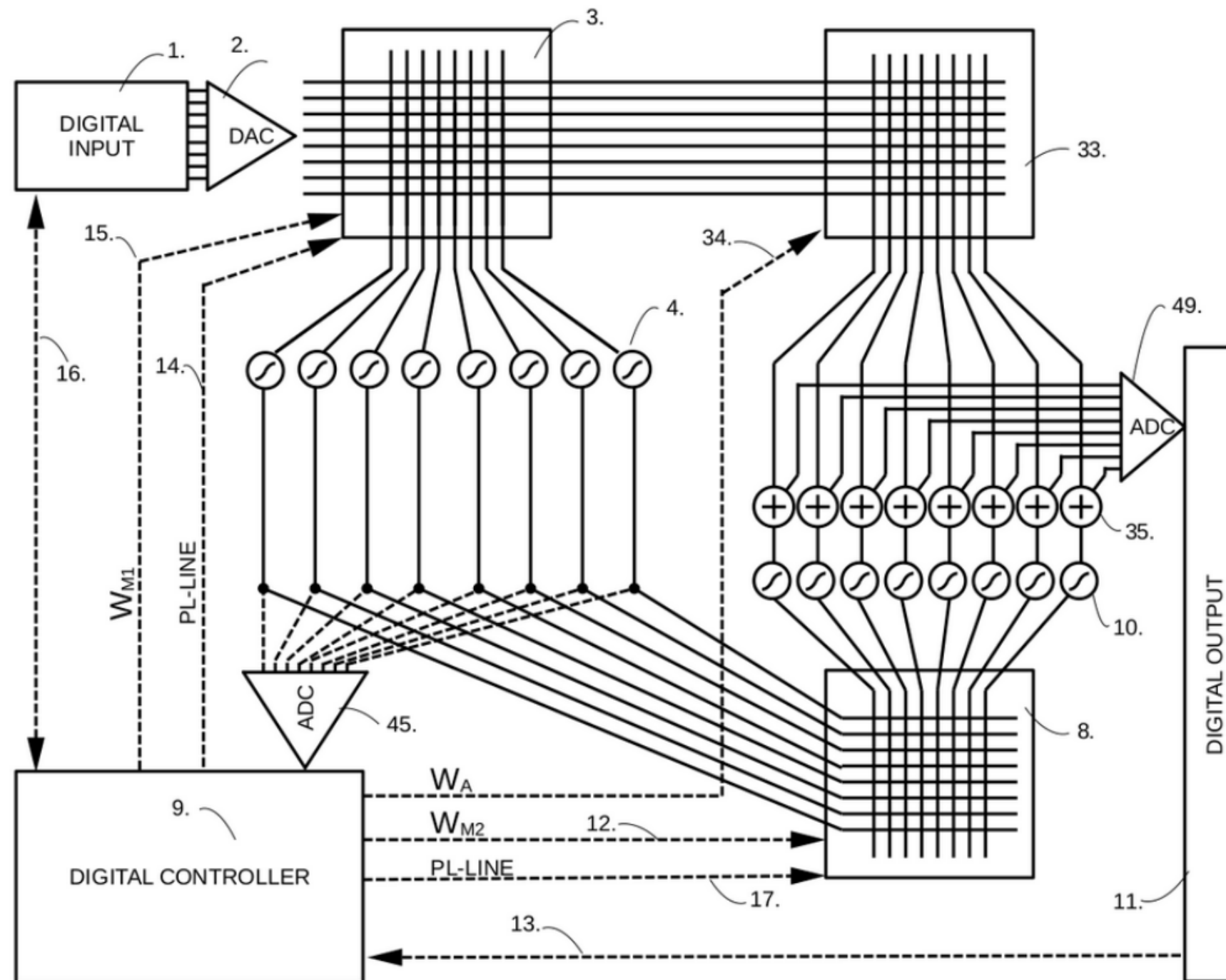
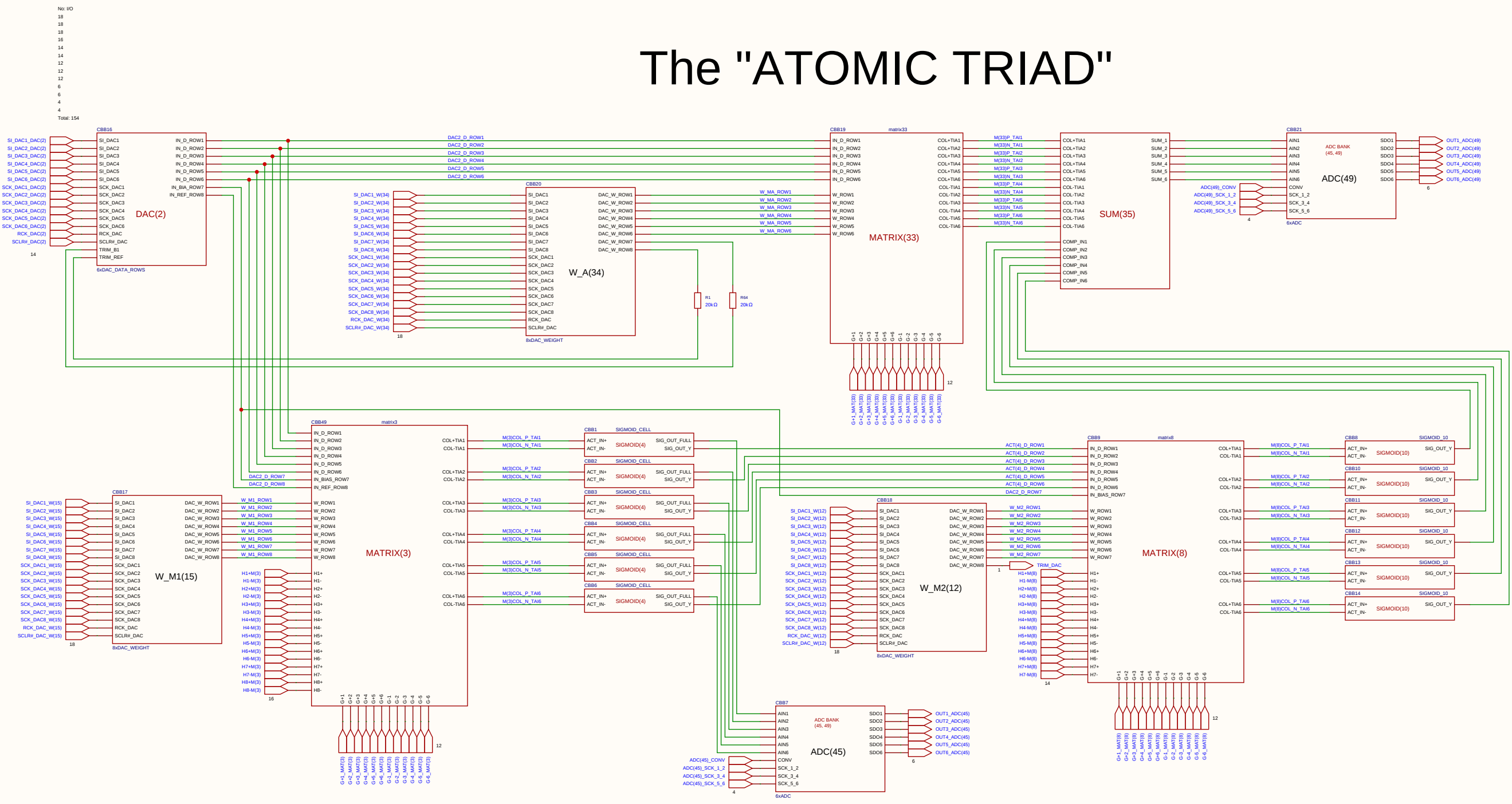


PRIORITY DATE: 2026-06-03



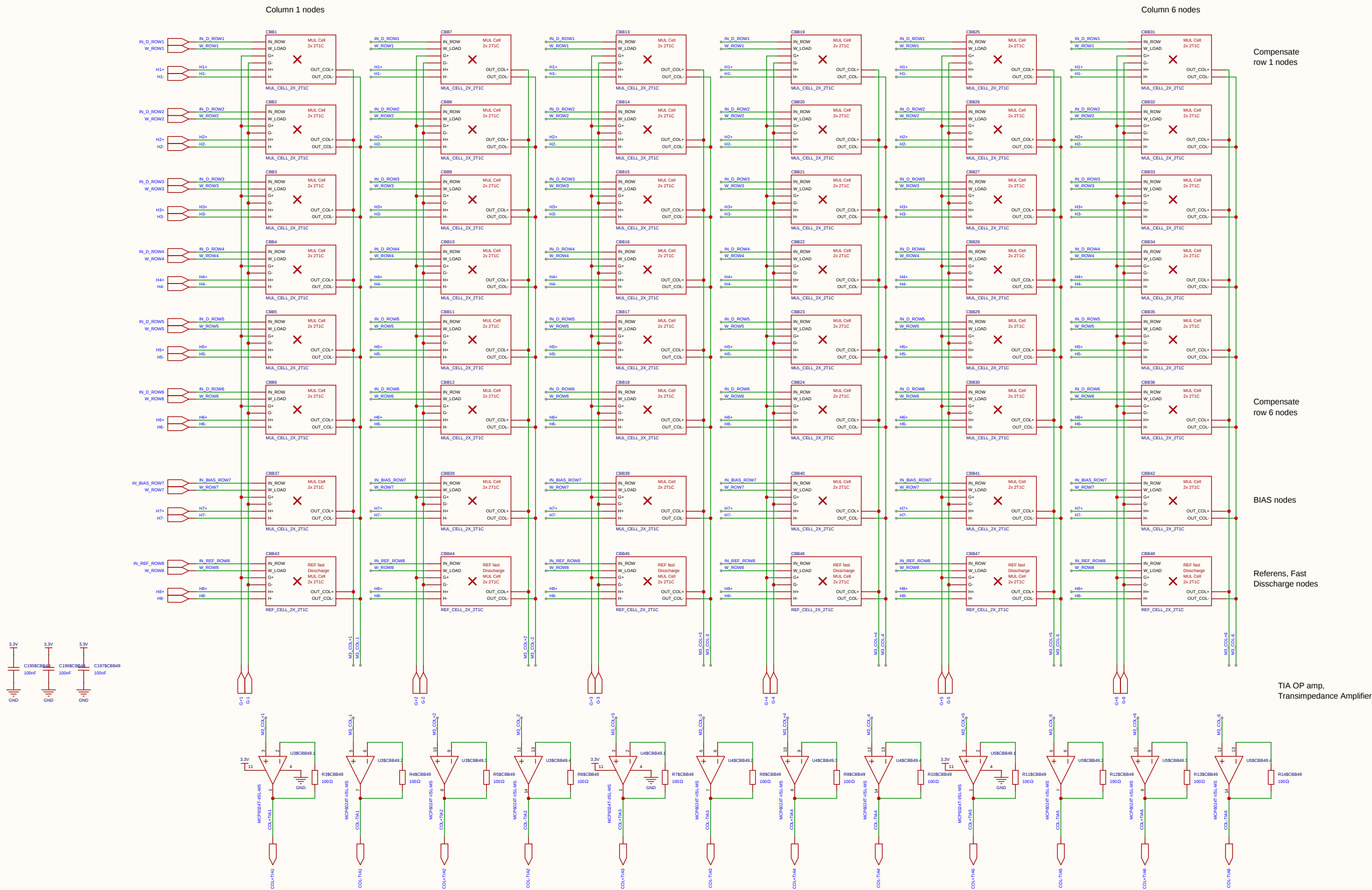
Schematic	Inverted_MAML			Create at	2026-06-19
				Update at	2026-06-19
Board	Analog_Matrix			Page	cover
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 3	
		V1.0	A4	EasyEDA.com	

The "ATOMIC TRIAD"



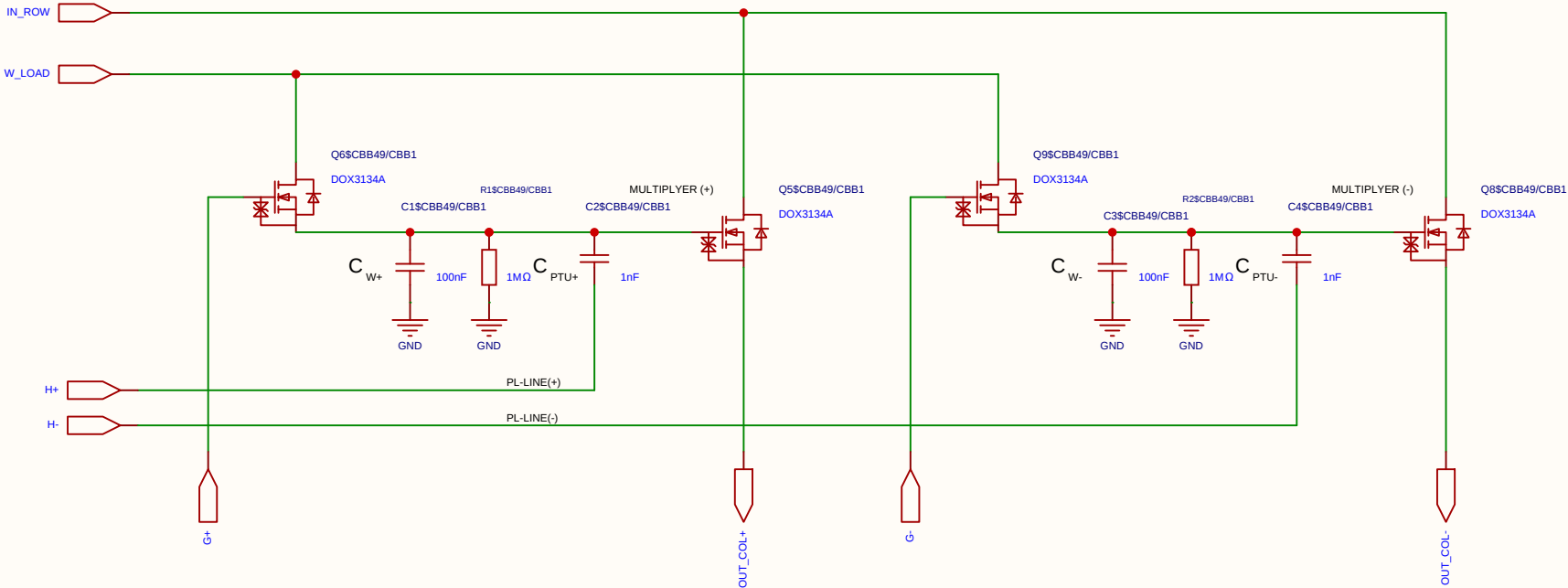
Schematic	Inverted_MAML			Create at	2026-06-25
				Update at	2026-06-27
Board	Analog_Matrix			Page	Atomic_Triad
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 2 Total 3	
		V1.0	A4	EasyEDA.com	

$$V_{ut_TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$



Schematic	matrix3				Create at	2026-06-25
					Update at	2026-06-27
Board					Page	matx3
Drawn		Inverted_MAML_6x6				
Reviewed						
		Version	Size	Page 1 Total 1		
		V1.0	A4	EasyEDA.com		

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

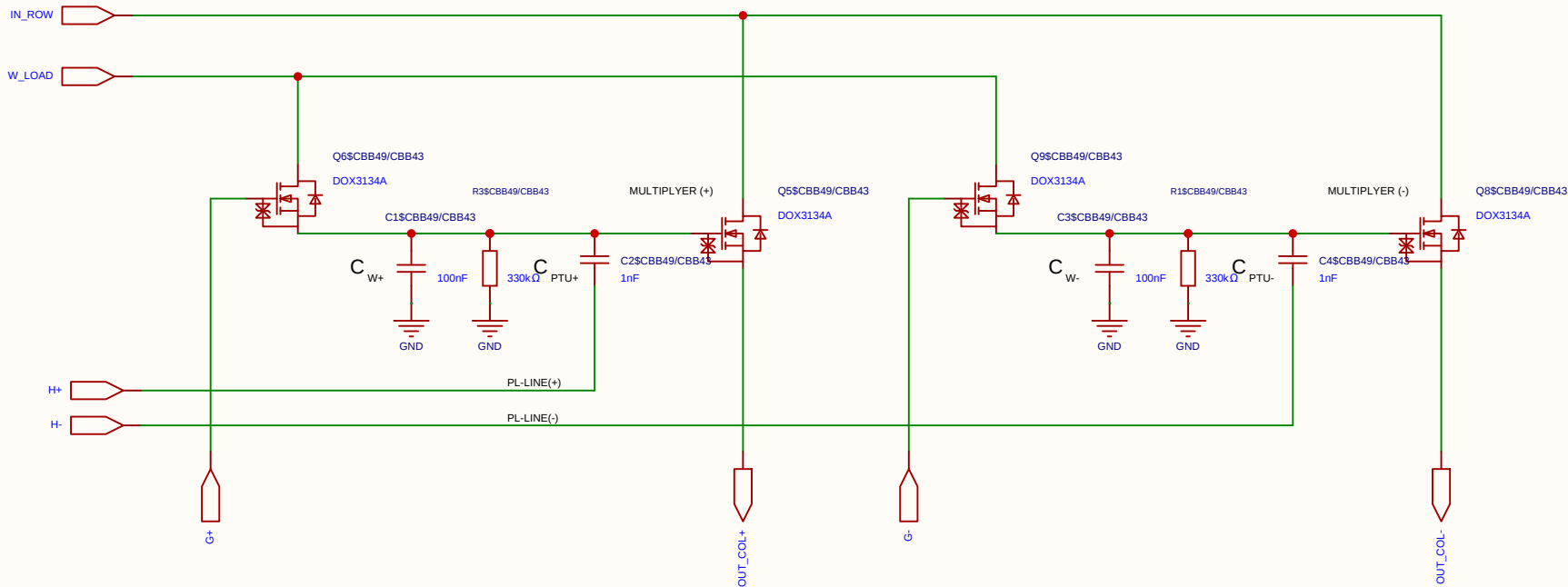
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

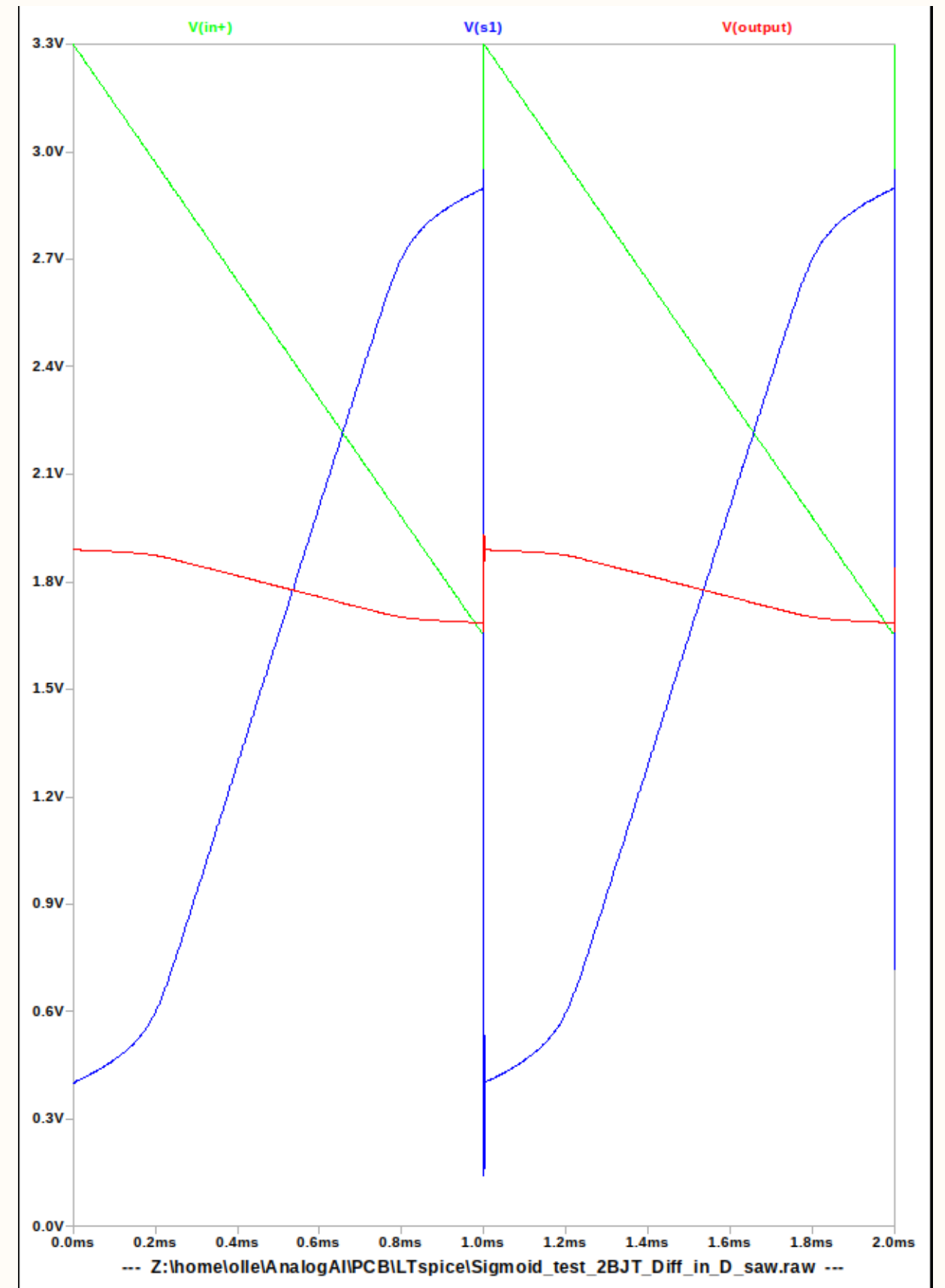
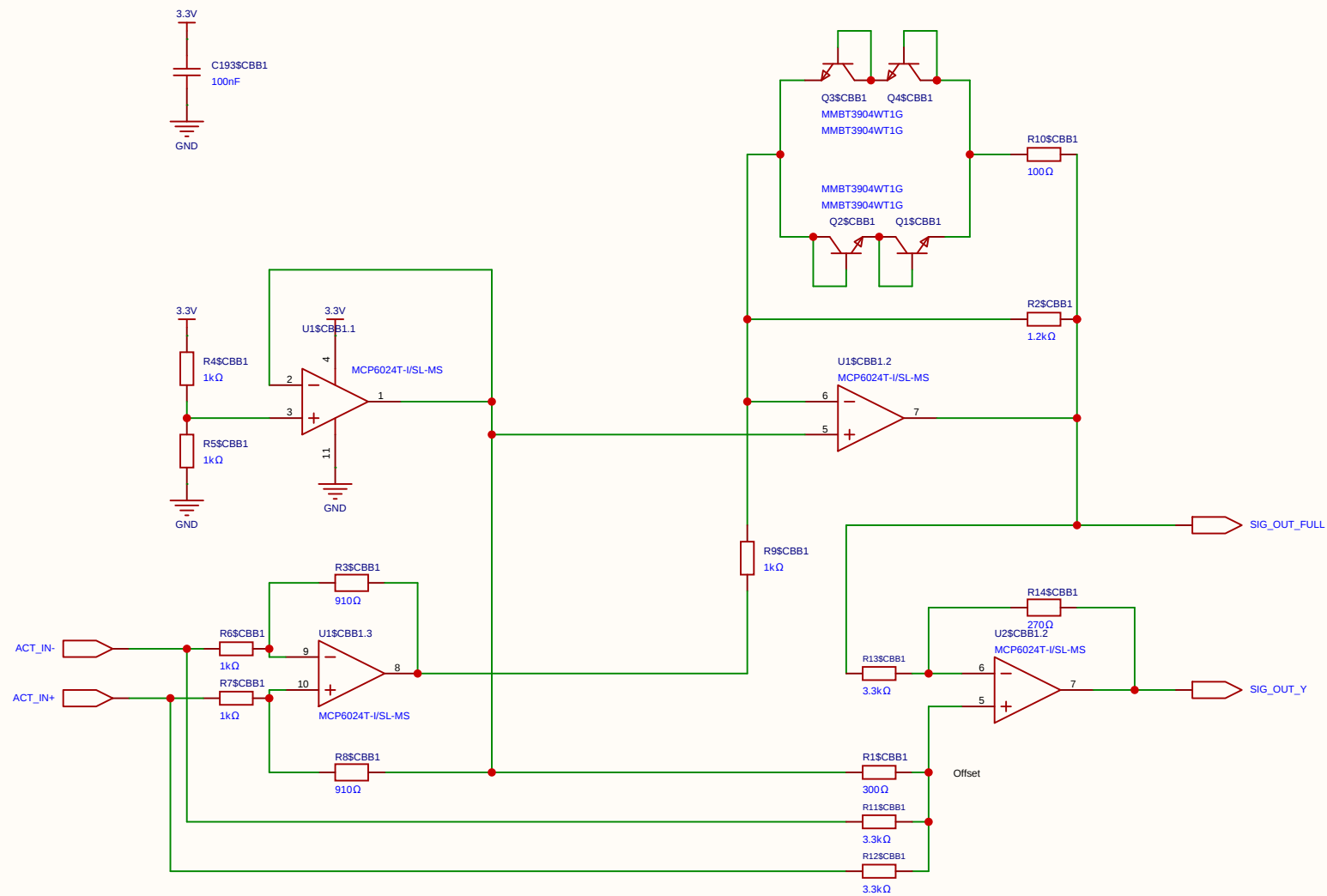
STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

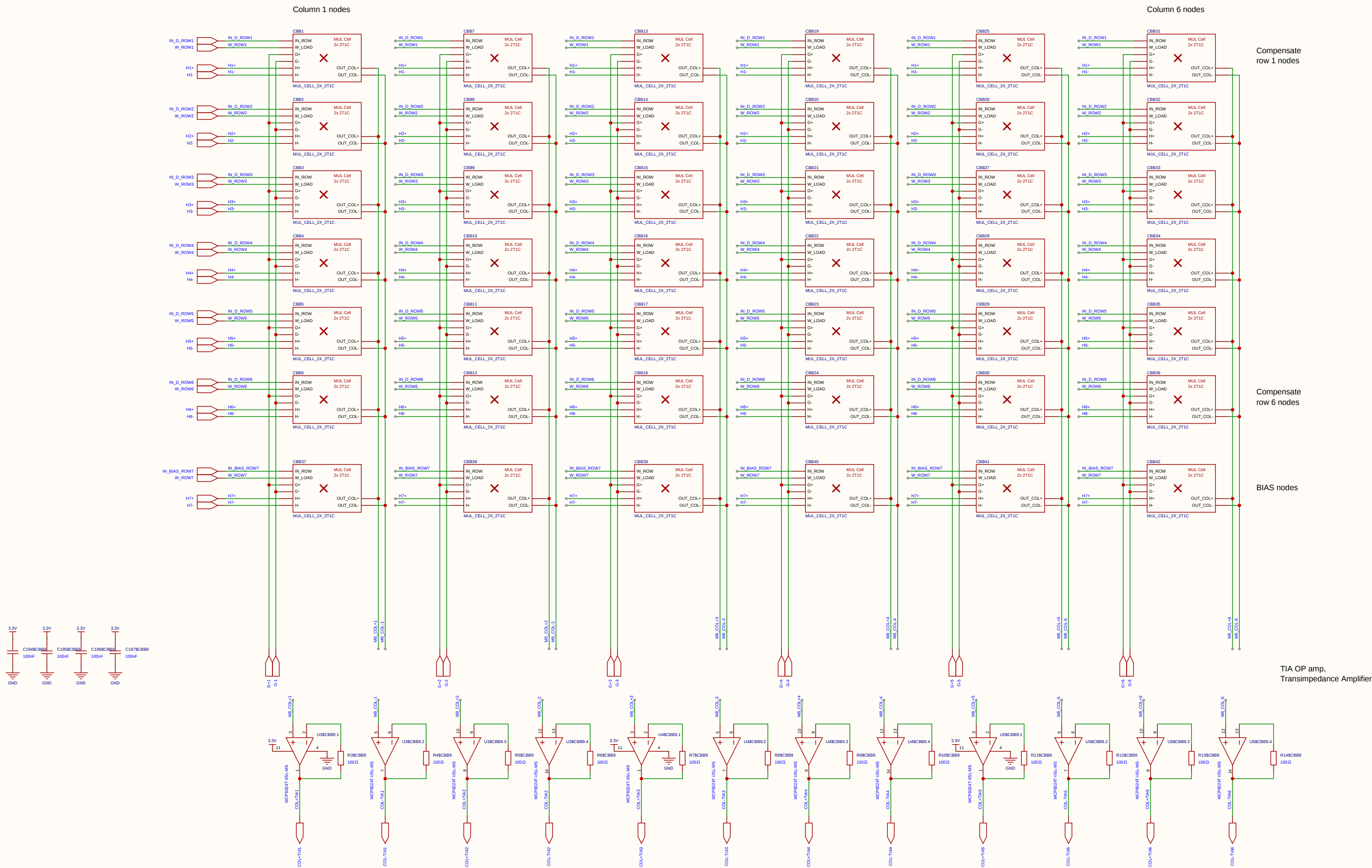
COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i . Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i . Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i . Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C			Create at	2026-06-19
				Update at	2026-06-30
Board				Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	



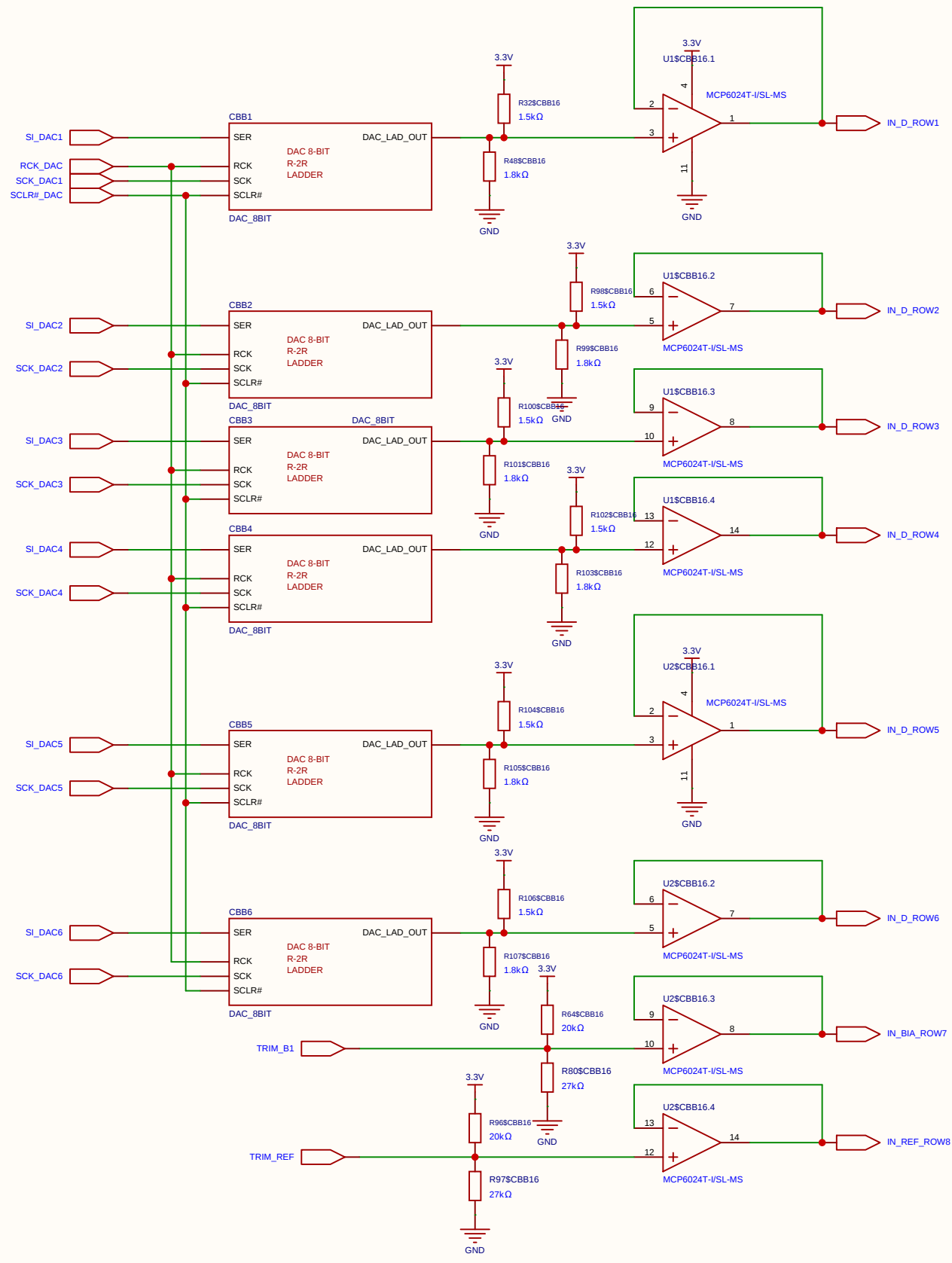
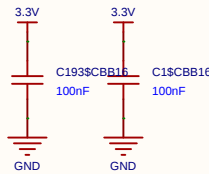
Schematic	SIGMOID_CELL			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	SIMOID(4)
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

$$V_{ut_TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$



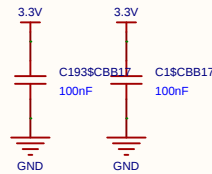
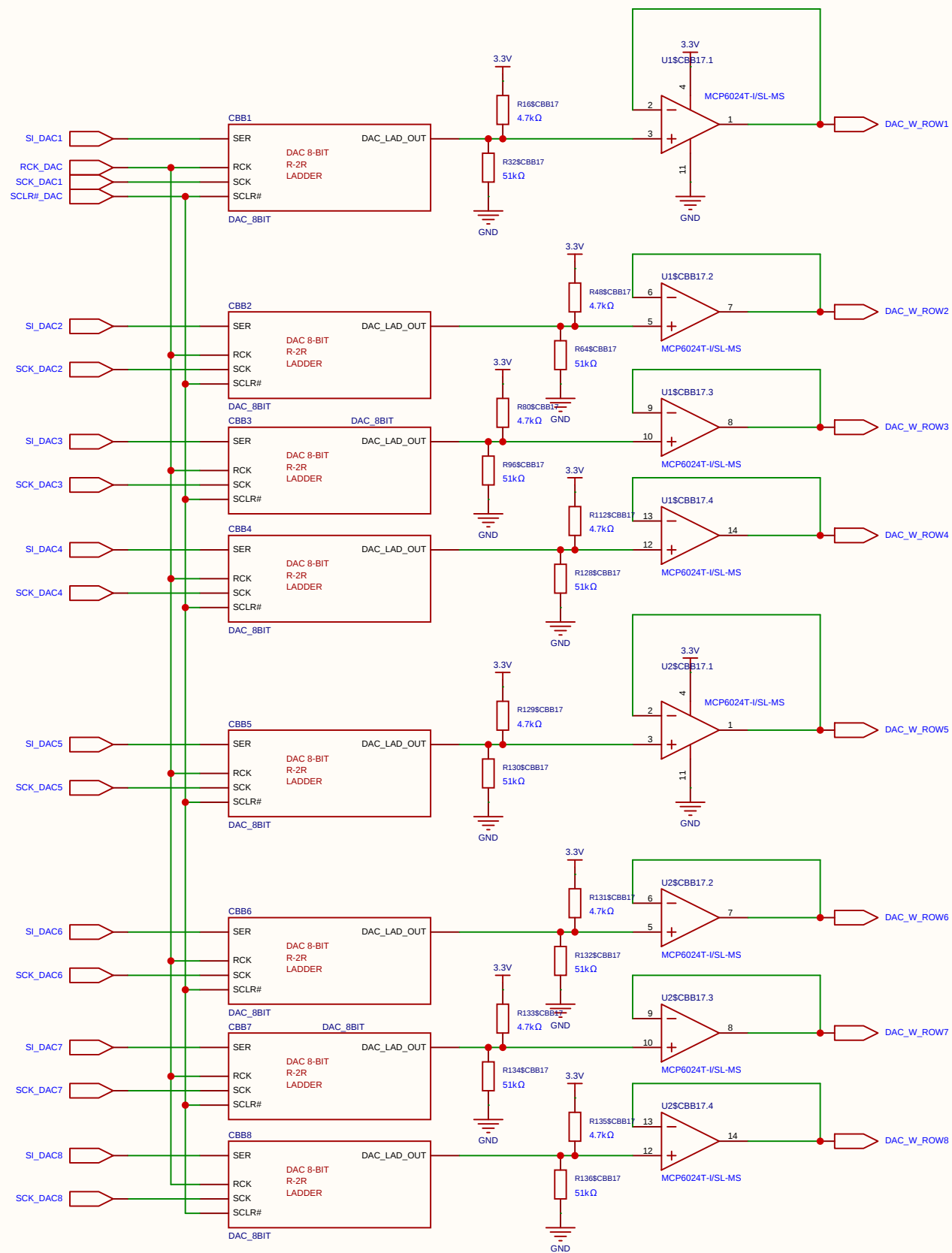
Schematic	matrix8			Create at	2026-06-25	
				Update at	2026-06-27	
Board	Page					matx8
Drawn		Inverted_MAML_6x6				
Reviewed						
		Version	Size	Page 1 Total 1		
		V1.0	A4	EasyEDA.com		

Voltages Data swing (Row / V_{ds}) 250 mV Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.



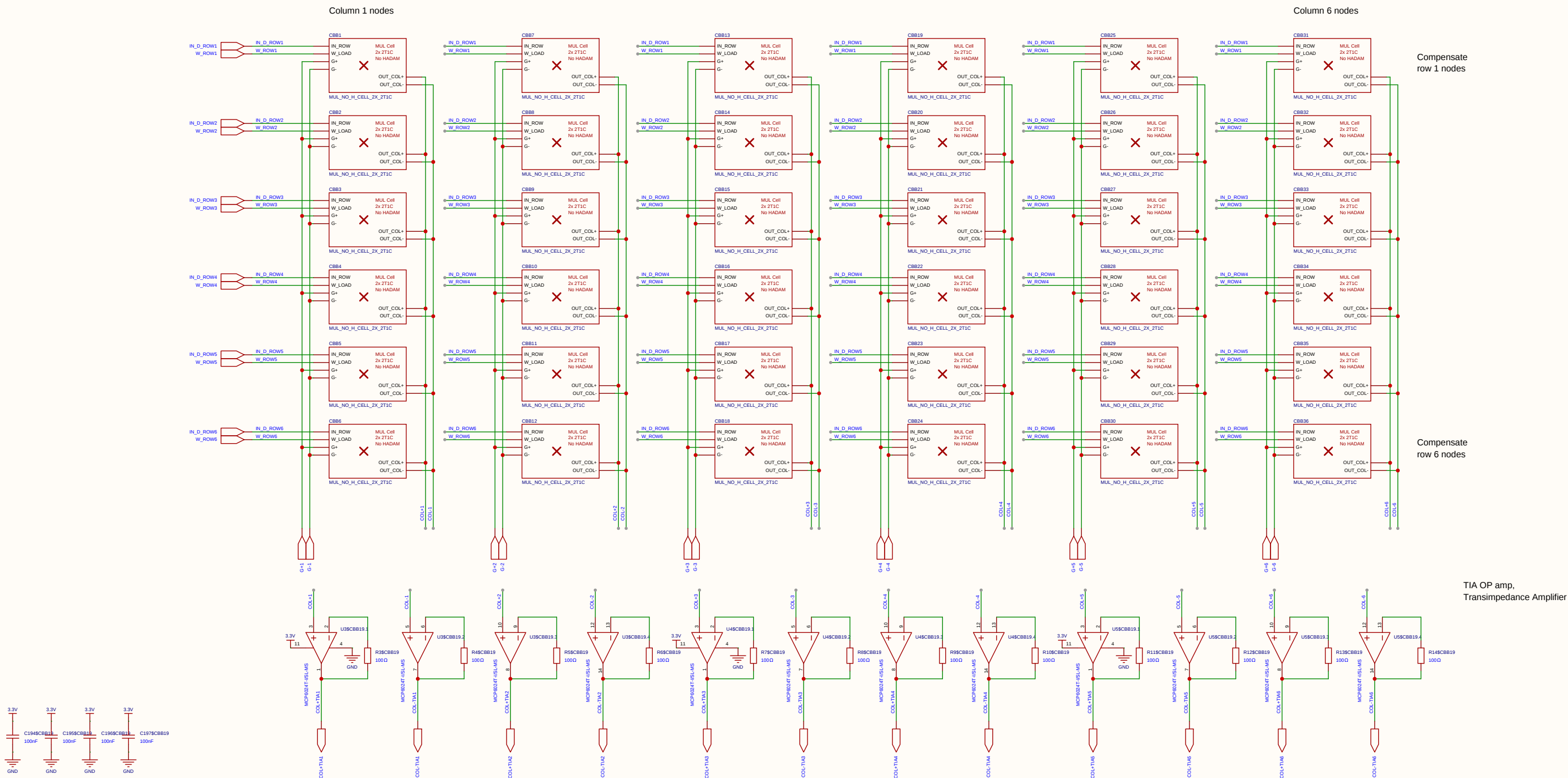
Schematic	6xDAC_DATA_ROWS			Create at	2026-06-20
				Update at	2026-06-20
Board				Page	6xDAC_8-bit
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Weight swing (Gate / V_{gs}) **1.0 V** Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).



Schematic	8xDAC_WEIGHT		Create at	2026-06-19
			Update at	2026-06-30
Board			Page	8xDAC_8-bit
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
		V1.0	A4	EasyEDA.com

$$V_{ut_TIA} = 1.65\text{ V} - (5\text{ mA} \times 100\ \Omega) = 1.65\text{ V} - 0.5\text{ V} = 1.15\text{ V}$$

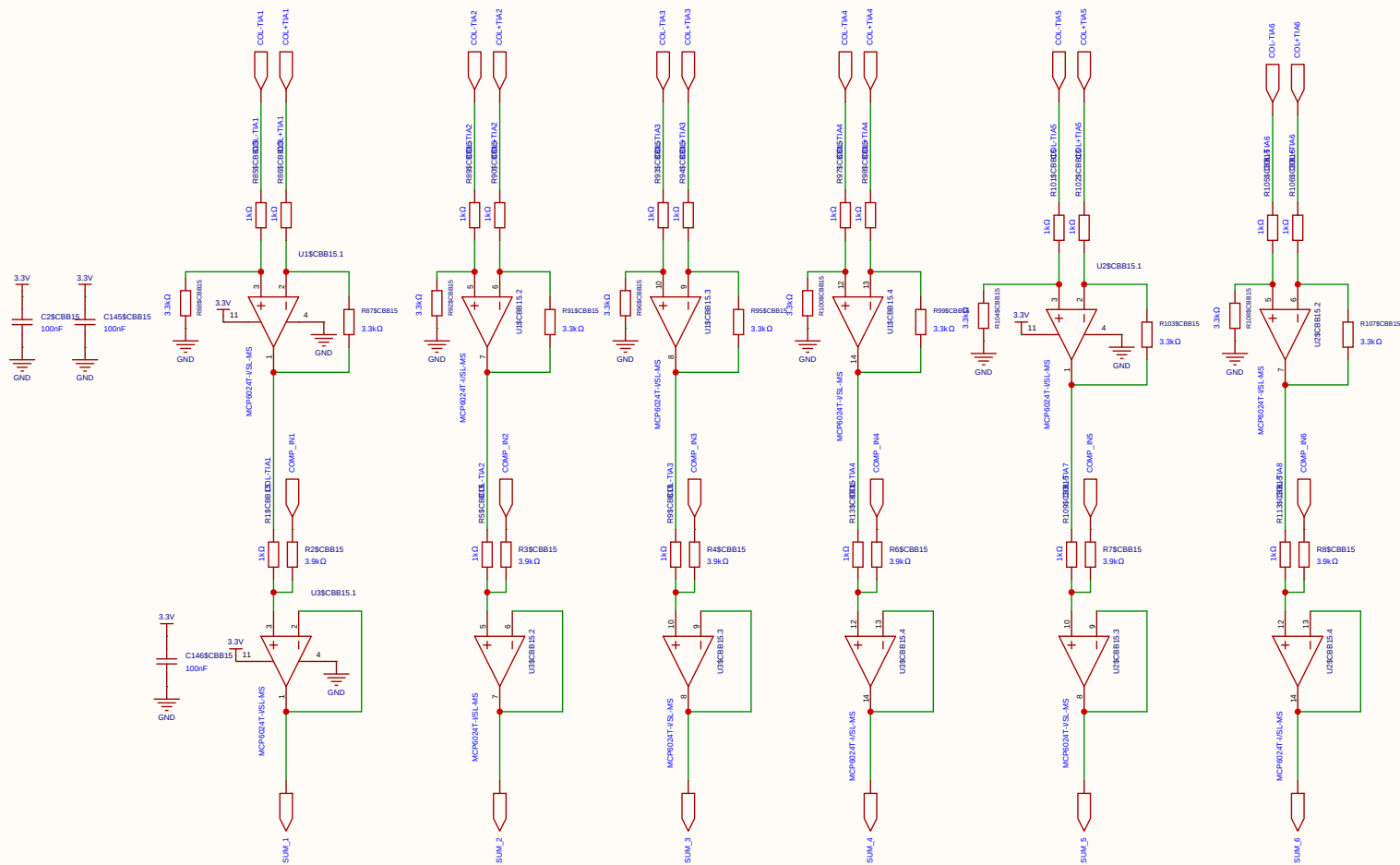


Schematic	matrix33			Create at	2026-06-25
Board				Update at	2026-06-28
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

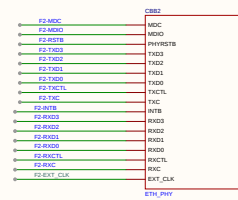
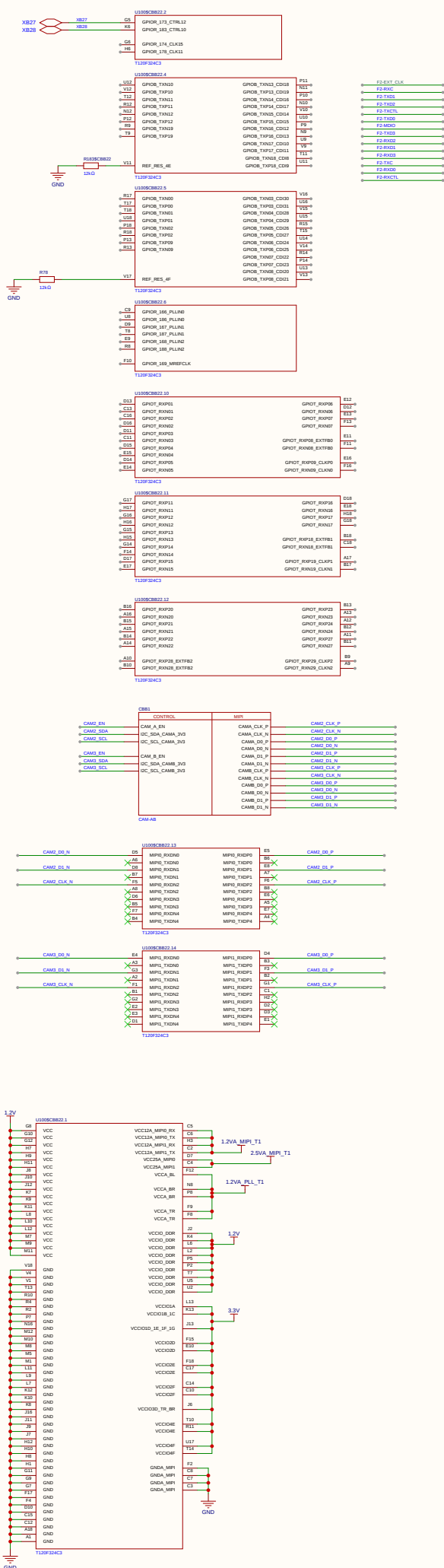
Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-20
				Update at	2026-06-28
Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

• **Calibration Example:** To match a 3.3 V ADC, the resistors are configured for a Gain of 6.6×.

$V_{ADC} = 6.6 \times (1.65 \text{ V} - V_{out_TIA})$

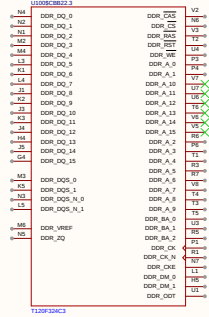
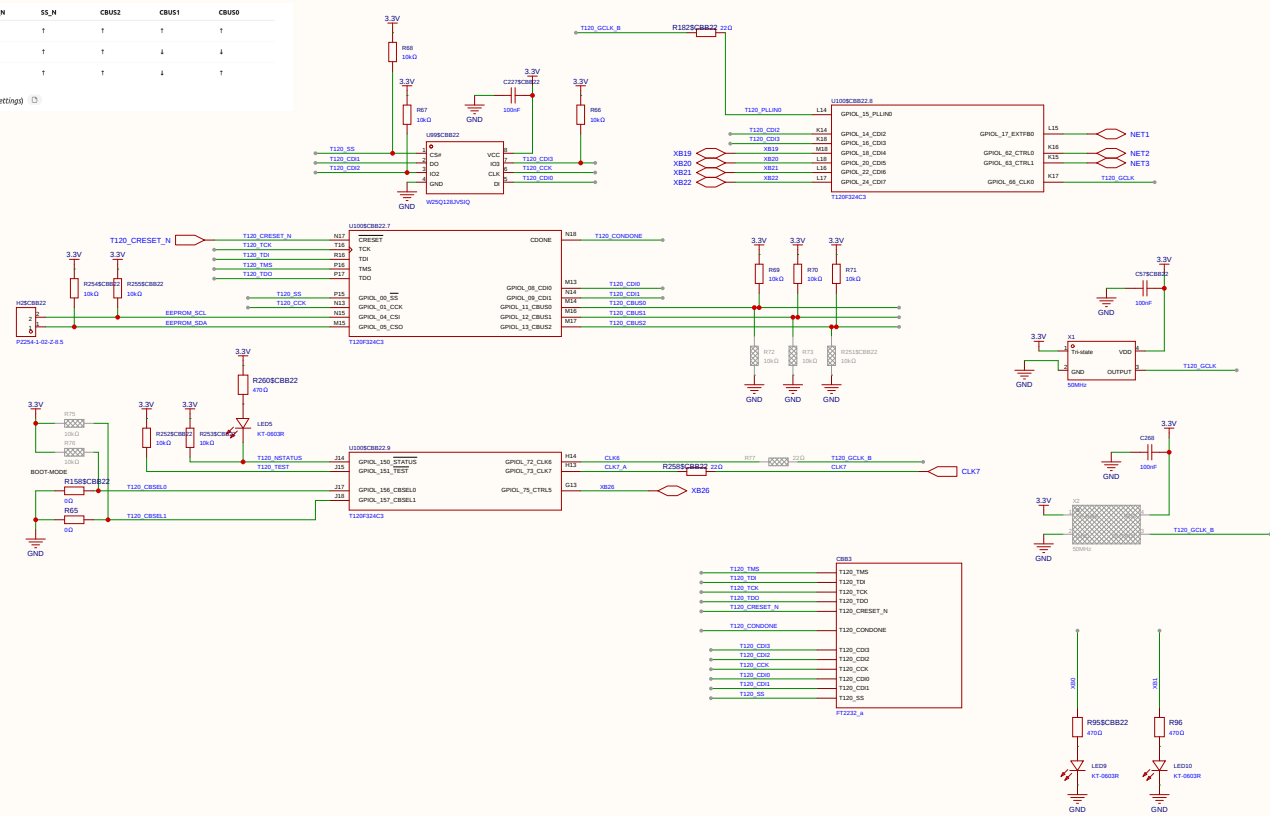


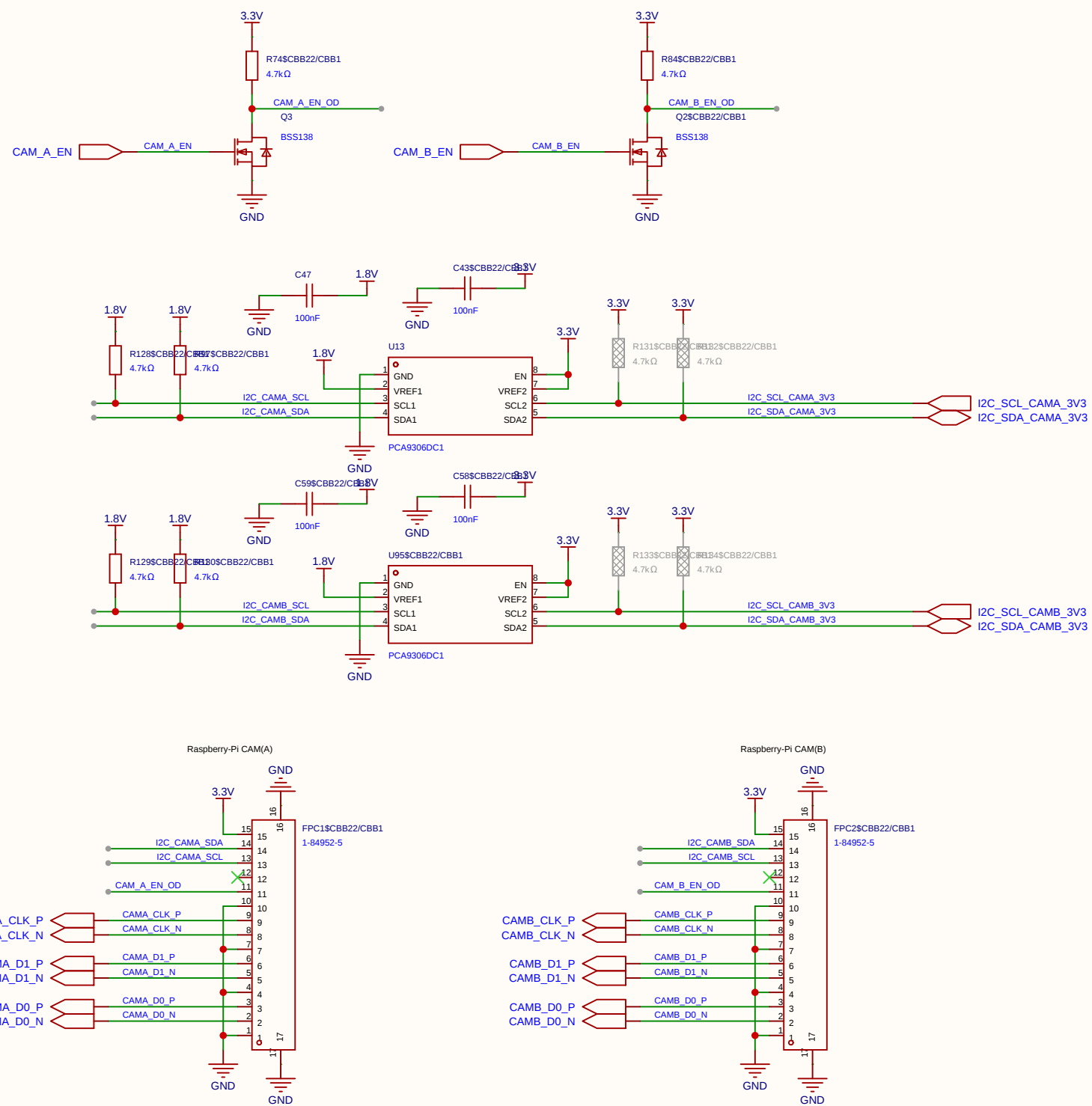
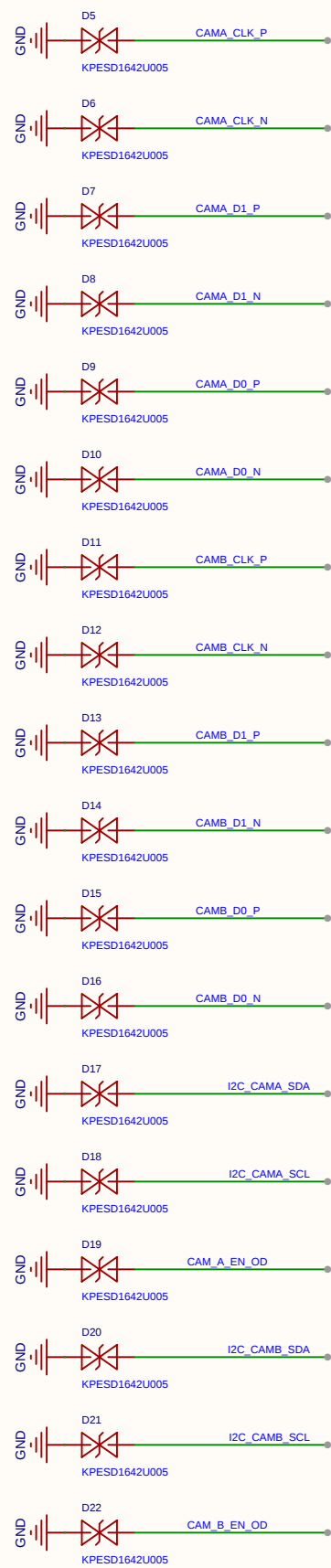
Schematic	SUM35			Create at	2026-06-19
Board				Update at	2026-06-19
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	



With	TEST_N	SS_N	CBUS2	CBUS1	CBUS0
x1	1	1	1	1	1
x2	1	1	1	1	1
x4	1	1	1	1	1

(Table 5. SPI Hardware Settings)





Schematic	CAM-AB			Create at	2026-06-25
				Update at	2026-06-25
Board				Page	P1
Drawn		Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
		V1.0	A4	EasyEDA.com	

